SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 1: Project Scope And Direction

Date and time	Wed 2 Sep 2026
Type	Supervisor
Location	In person (Dr. Panos's office, D04208)
Attendees	Dr. Panos, Khaled, Zain, Ayesha, Tehsin, Zayed

Agenda

1. Project scope and technical challenges
2. Data sources
3. Datasets and labelling
4. Deliverables, SDP I / SDP II split and feasibility
5. Team coordination and allocation status

Member updates

Who	Update
Zain	Represented the team in the scoping discussion

Discussion

1. Project scope and technical challenges

- Goal: detect meaningful change between satellite images of the same Abu Dhabi area.
- Main problems: aligning images, segmentation, and separating real ground change from cloud, lighting and other acquisition differences.
- Mangroves, urban edge and coast were debated; Dr. Panos recommended a realistic case study centred first on mangroves.

2. Data sources

- Sentinel-2 (optical) and Sentinel-1 (radar). Radar works through cloud and at night, but aligning optical and radar at different resolutions is hard.
- Abu Dhabi imagery from Sentinel via Google Earth Engine; local UAE imagery as a fallback.

3. Datasets and labelling

- Start module development on co-registered, densely labelled datasets such as S2Looking and LEVIR.
- In parallel, work on Abu Dhabi data, turning machine-generated regions into human-readable labels.
- Dr. Panos suggested using texture and colour, and fine-tuning a pretrained SAM-based model.

4. Deliverables and feasibility

- Multi-sensor fusion, co-registration and varying conditions make this substantial even for five people.

- One-year plan: SDP I = tools, datasets, algorithm tests and a preliminary study; SDP II = system integration.
- Test each module, integrate where feasible, and report limitations.

5. Team coordination

- Meet Dr. Panos weekly; meet internally more often; at least two people per task.
- Use Dr. Panos's Outlook calendar to schedule.
- Dr. Jamal announces allocations; wait until the following week before substantive work.

Decisions

- Mangrove change detection is the committed deliverable; other areas only if capacity allows.
- Sentinel-1 and Sentinel-2 are the primary data sources.
- SDP I covers modules and the Abu Dhabi evaluation set; SDP II covers integration.
- At least two people per task; weekly supervisor meetings.

Action items

Who	What	Due
Team	Define technical accuracy targets for modules and the integrated system	Not set
Team	Prepare the Abu Dhabi evaluation set	During SDP I
Team	Identify co-registered, densely labelled datasets for initial development	Not set
Team	Investigate Abu Dhabi imagery sources (Sentinel, Earth Engine, local UAE)	Not set
Team	After allocation, meet Dr. Panos to break the project into tasks; assign ≥ 2 people per task with deadlines	After allocation
Zain	Arrange a recurring meeting time	After allocation

Next meeting

After project allocation is confirmed (held Mon 14 Sep).